Stay Flexible, Maneuver Less:
Behavioral Signatures of Efficient Problem Solving in Constraint Satisfaction Puzzles

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Abstract

What distinguishes efficient problem solvers from those who struggle? We analyzed behavioral trajectories from 221 nonogram puzzle attempts to understand how solving strategies relate to efficiency. Our central finding is that students whose constraint satisfaction progress occurred later in the solving process travel significantly less total distance than those whose progress occurred early ($r = -0.52$, Cohen's $d = 1.14$). We interpret early progress as reflecting a greedy approach (completing constraints sequentially) and late progress as reflecting a distributed approach (working across many constraints before completing any). Notably, greedy solvers showed higher variance: some finished efficiently when early commitments proved correct, while others became trapped in extensive backtracking. Distributed solvers were consistently efficient. The effect was robust across both puzzle variants and all grade levels examined.

1 Introduction

Complex problems often require solvers to coordinate multiple interacting constraints. A common challenge is that locally sensible decisions can create global conflicts. Completing one part of a problem may inadvertently make another part unsolvable. Some solvers navigate this challenge gracefully, while others struggle with repeated backtracking. Understanding what behavioral patterns distinguish efficient solvers has implications for education, training, and cognitive science.

Constraint satisfaction problems offer a lens for studying this question. In these problems, a solver must find a configuration that simultaneously satisfies multiple requirements. Examples include scheduling, resource allocation, and logic puzzles.

1.1 Nonograms as a Model Task

Nonograms are spatial logic puzzles played on a grid (Batenburg & Kosters, 2012). Each row and column carries a numerical hint specifying how many cells should be filled. The solver must determine which cells to fill such that all hints are satisfied simultaneously.

What makes nonograms valuable for studying problem solving is their constraint structure. Each cell participates in two constraints: its row hint and its column hint. Filling a cell advances progress on both, but overfilling a row to satisfy a column creates a conflict requiring correction. Success demands coordinating progress across all constraints rather than completing them sequentially.

The puzzles used in this study specify row and column hints as sums, meaning multiple valid solutions exist. This design allows us to study strategy rather than mere accuracy.

1.2 Research Question

We examined whether the timing of constraint satisfaction predicts efficiency. Specifically, we asked: do students whose progress occurs earlier tend to backtrack more or less than those whose progress occurs later? We interpret early progress as reflecting a greedy approach and late progress as reflecting a distributed approach, though Progress COM measures timing rather than strategy directly.

2 Data and Measures

2.1 Participants and Task

We analyzed behavioral data from the CTY Dynamic Spatial Assessment project. Students solved two 10×10 nonograms (Figure 1) with complete action logging. There was no time limit and students received no score feedback during solving.

The sample comprised 131 students who completed 221 puzzle attempts (Table 1). Fifty incomplete attempts were excluded. Most students (90 of 131) completed both puzzles; the remaining 41 completed one. The two puzzles were balanced with 117 attempts on Puzzle 3 and 104 on Puzzle 4.

Grade	3	4	5	6	7	8	9	10	11	12
n	4	2	4	10	47	26	61	34	29	4

Table 1: Distribution of completed attempts by grade. The majority of attempts (197 of 221) were from students in grades 7–11.

Within-student correlations across puzzles were weak ($r = 0.22$ for Progress COM, $r = 0.07$ for distance), suggesting students did not behave consistently across puzzles. The main effect remained significant when accounting for within-student clustering using mixed effects regression ($\beta = -164$, $p < 0.001$).

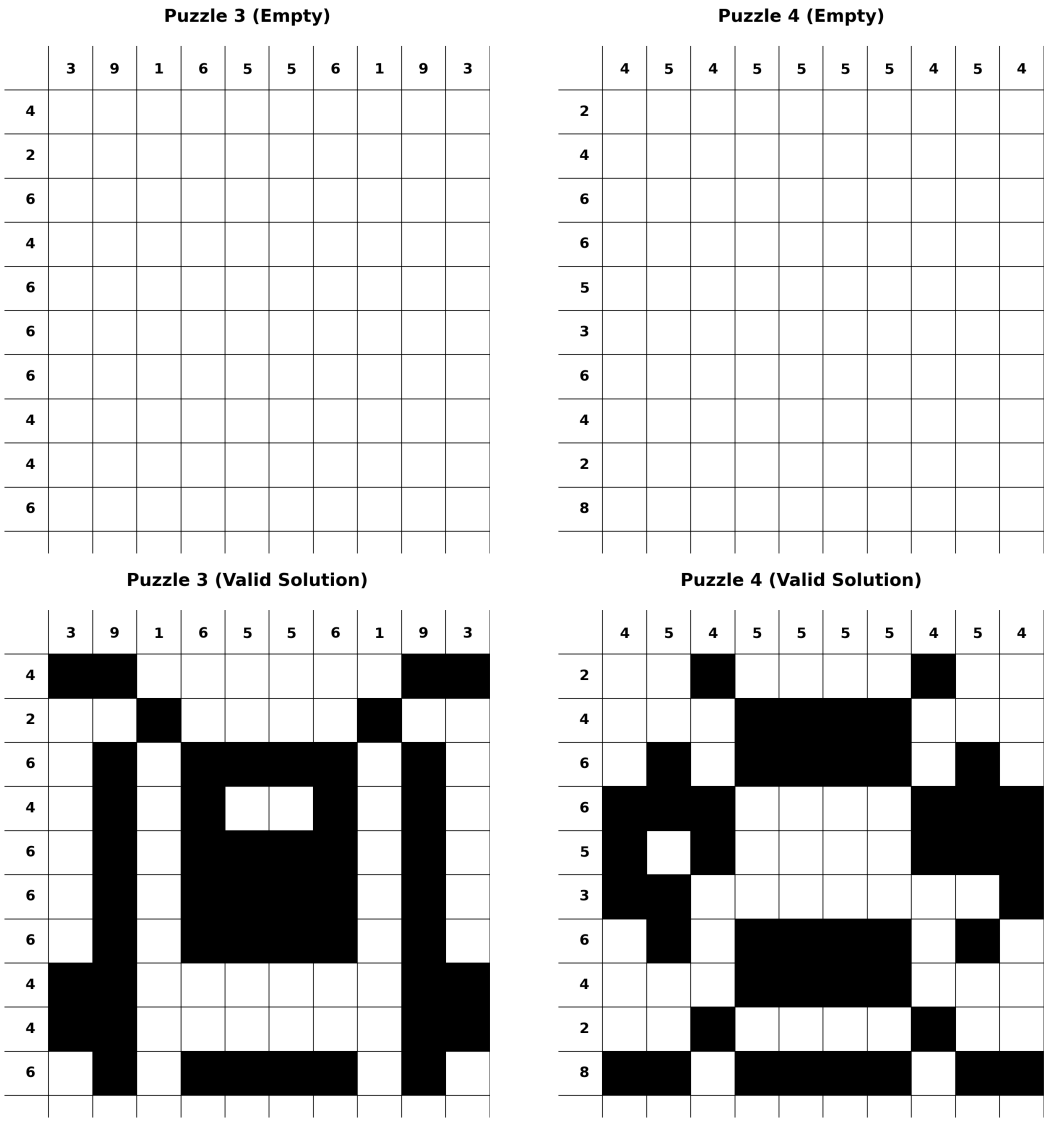

Figure 1: The two nonogram puzzles used in this study. Top: empty grids with row hints (left) and column hints (top). Bottom: one valid solution for each puzzle satisfying all constraints.

For analyses involving correction timing and concentration, we restricted to 178 attempts with at least five corrections to ensure meaningful measurement of when and how corrections clustered.

2.2 Behavioral Measures

The logging system recorded every action: each cell filled or unfilled, along with timestamps. Reversions include unfill and undo actions. Redo actions were not counted as reversions since they restore previous fills. Resets were included in reversion counts but assigned zero distance to avoid inflation.

2.2.1 Efficiency Measures

We considered five candidate measures of efficiency: total time, total actions, total distance (cumulative Manhattan distance across all actions), number of reversions, and reversion percentage (reversions divided by total actions). We focus on total distance because it captures overall movement cost and correlates strongly with progress timing.

2.2.2 Progress Timing

To measure when progress occurred during solving, we divided each attempt's action sequence into five equal bins and computed how much constraint satisfaction increased within each bin. Progress COM (center of mass) is the weighted average bin, where weights are the proportion of total progress made in each bin:

$$\text{Progress COM} = \sum_{b=0}^{4} b \cdot p_b$$

where p_b is the fraction of total line score increase that occurred in bin b. Values range from 0 (all progress in the first bin) to 4 (all progress in the last bin).

Low Progress COM means most line completions happened early, consistent with a greedy approach of completing constraints as you go. High Progress COM means completions were deferred until later, consistent with a distributed approach of exploring before committing to any constraint.

Correction COM applies the same computation to correction actions, measuring when corrections occurred. Correction concentration is measured using the Gini coefficient of the correction PMF. The Gini coefficient is a measure of inequality commonly used in economics; here it captures how unevenly corrections are distributed across bins:

$$G = \frac{2\sum_{i=1}^{n} i \cdot p_{(i)}}{n\sum_{i=1}^{n} p_{(i)}} - \frac{n+1}{n}$$

where $p_{(i)}$ is the i-th value of the PMF sorted in ascending order. Values range from 0 (corrections evenly spread across all bins) to 1 (all corrections in a single bin).

3 Results

3.1 Main Finding: Distributed Approach Predicts Efficiency

Students whose progress occurred later in the solving process traveled less total distance. The correlation between Progress COM and total distance was $r = -0.52$ ($p < 0.001$). Figure 2 displays this relationship.

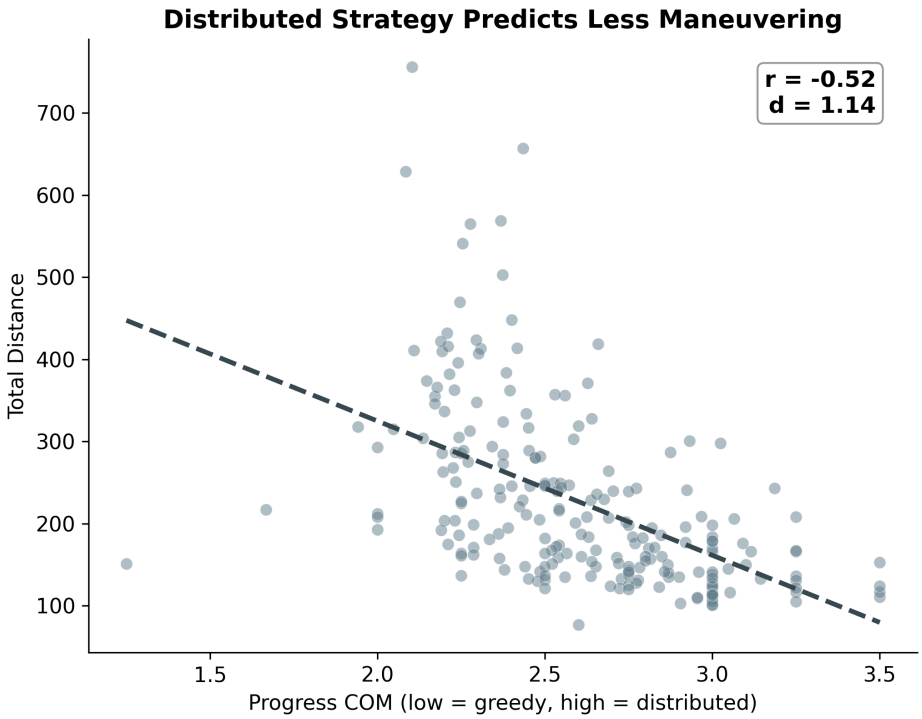

Figure 2: Distributed approach predicts efficiency. Each point represents one puzzle attempt. Students with higher Progress COM (distributed approach) travel less total distance. Effect size: $r = -0.52$, $d = 1.14$.

Splitting students at the median Progress COM (2.56), those below the median averaged 281.9 distance units (SD = 122.1) while those above averaged 172.0 (SD = 62.4), yielding Cohen's $d = 1.14$, a very large effect.

3.2 Greedy Approach Shows Higher Variance

The correlation does not imply that a greedy approach always leads to poor outcomes. Comparing tertiles of Progress COM reveals an important asymmetry (Figure 3, Table 2).

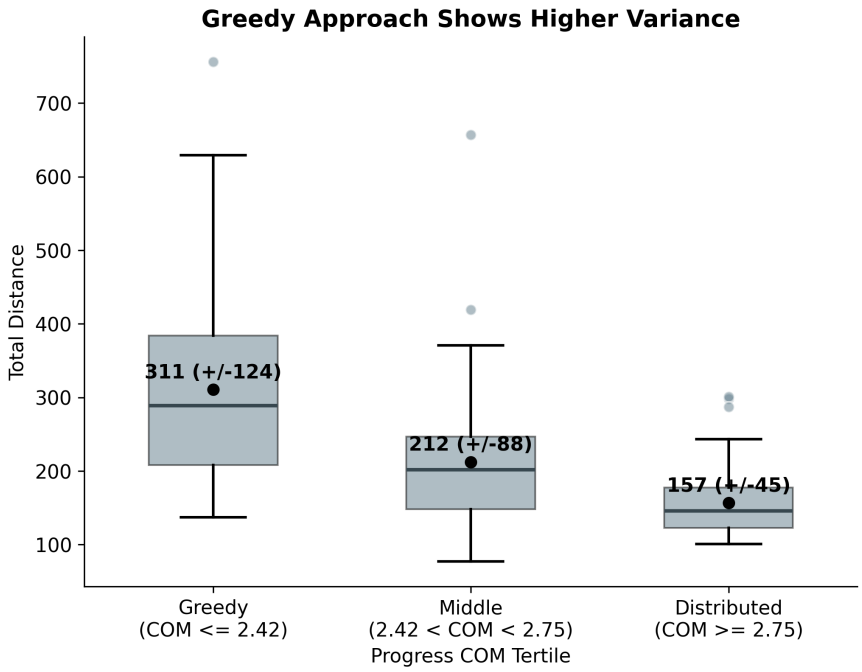

Figure 3: Greedy approach shows higher variance. Box plots show distance distribution by Progress COM tertile. Greedy solvers (low COM) have both higher mean distance and nearly three times the standard deviation of distributed solvers (high COM).

	Greedy ($\text{COM} \leq 2.42$)	Middle ($2.42 < \text{COM} < 2.75$)	Distributed ($\text{COM} \geq 2.75$)
Mean distance	311	212	157
SD distance	124	88	45

Table 2: Students with a greedy approach show higher variance in efficiency.

Students with a greedy approach had nearly three times the variance of those with a distributed approach (SD = 124 vs 45). Some greedy solvers finished efficiently, likely because their early commitments happened to be correct. Others became trapped in extensive backtracking. Students with a distributed approach were consistently efficient.

This pattern suggests that a greedy approach is a gamble: it can work out well, but it carries substantial risk of costly backtracking. A distributed approach is a safer, more reliable path to efficiency.

3.3 Consistency Across Efficiency Measures

Progress COM predicts multiple efficiency measures (Figure 4). Distance shows the strongest correlation ($r = -0.52$), followed by total actions ($r = -0.48$), reversions ($r = -0.43$), time ($r = -0.40$),

and reversion percentage ($r = -0.40$). Effect sizes follow a similar pattern, with distance showing the largest Cohen's d (1.14).

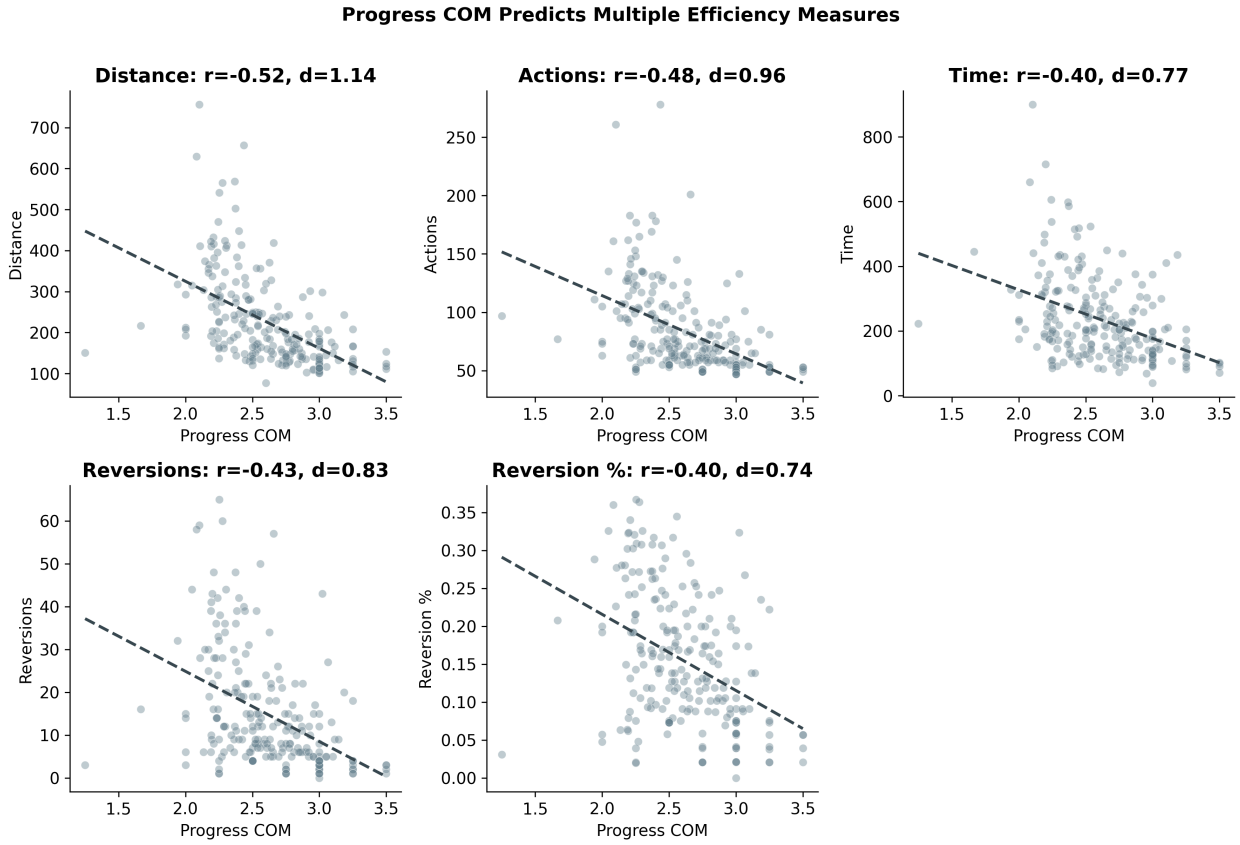

Figure 4: Progress COM predicts multiple efficiency measures. Each panel shows the relationship between Progress COM and one measure. Distance shows the strongest relationship.

3.4 Distribution of Progress Timing

Figure 5 shows the distribution of Progress COM across attempts. The distribution is unimodal and roughly symmetric (mean = 2.60, SD = 0.35), indicating continuous variation rather than discrete subgroups.

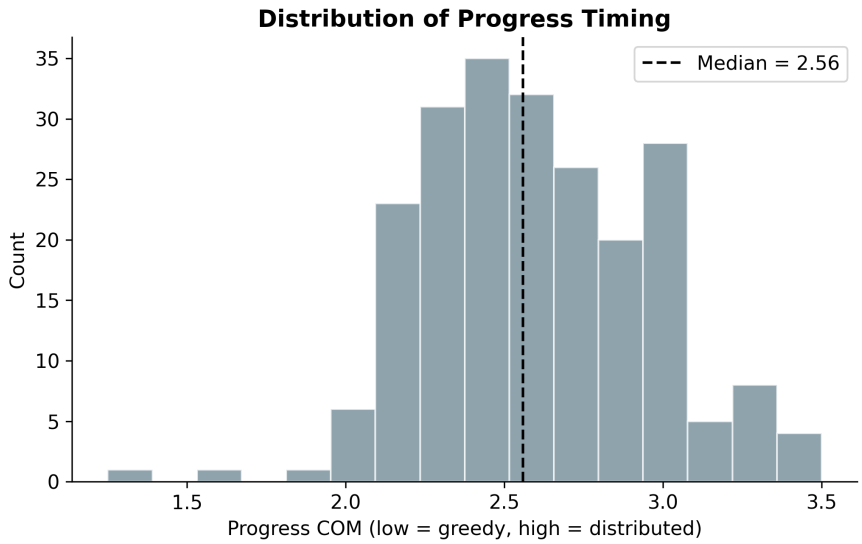

Figure 5: Distribution of Progress COM. Students vary continuously in when their progress occurs; there are no distinct subgroups.

3.5 Progress Timing and Correction Timing

Using the restricted sample of 178 attempts with at least five corrections, Progress COM and Correction COM were moderately correlated ($r = -0.38$; Figure 6). Students whose progress occurred late tended to make corrections earlier in the solving process.

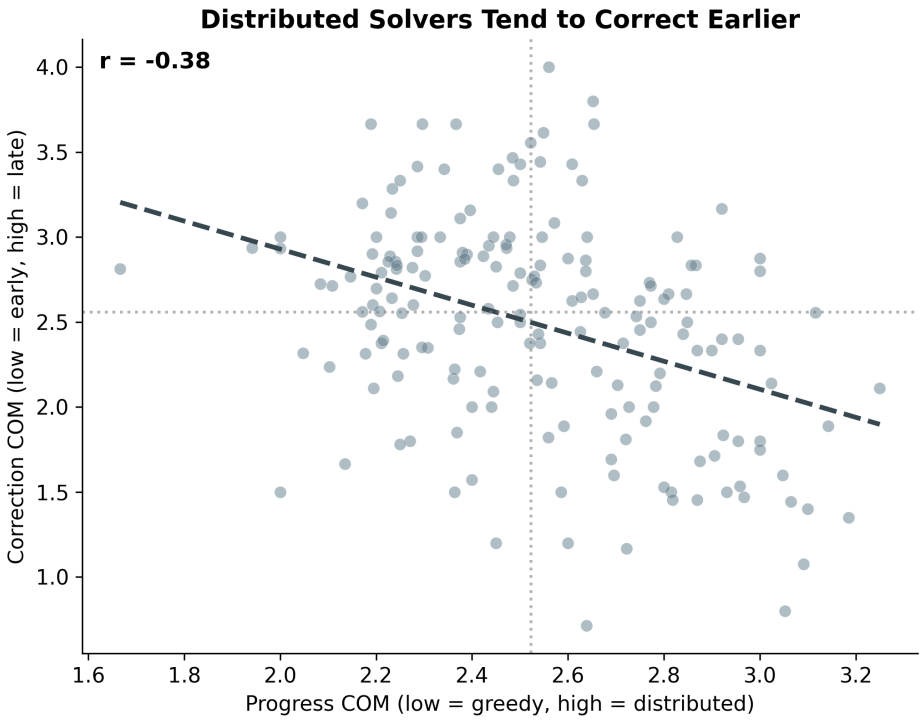

Figure 6: Distributed solvers correct earlier. Progress COM and Correction COM are negatively correlated ($r = -0.37$, $n = 178$).

3.6 Robustness Across Puzzles

The main effect replicated across both puzzles (Figure 7): $r = -0.57$ for Puzzle 3 ($n = 117$) and $r = -0.46$ for Puzzle 4 ($n = 104$). The correlation did not differ significantly between puzzles (interaction $p = 0.32$).

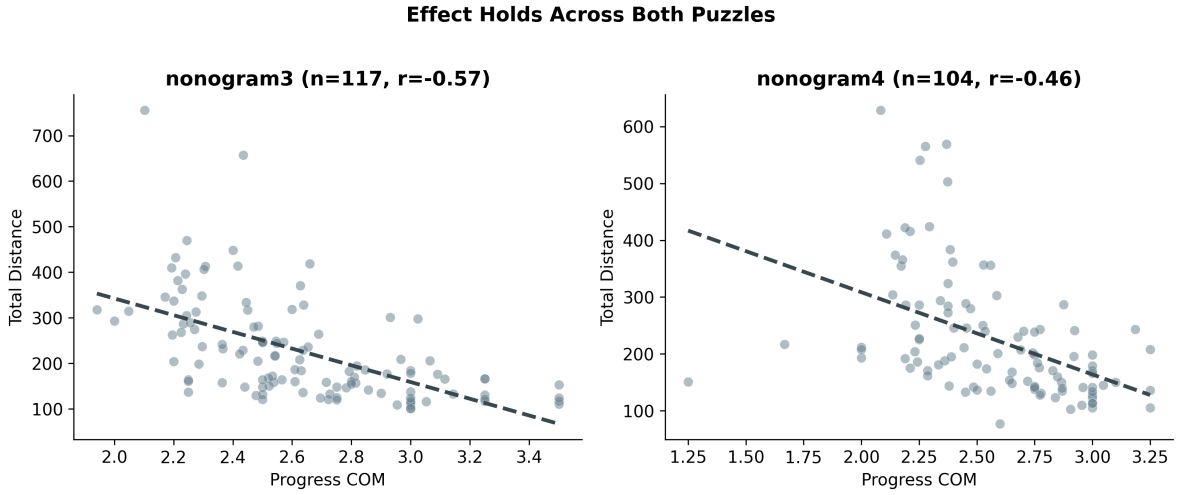

Figure 7: The effect holds for both puzzles despite their different constraint structures.

3.7 Robustness Across Grades

The effect held across all grade levels with sufficient sample sizes (Figure 8). Correlations were negative and consistent across grades 6 to 11, all in the same direction. Grade did not moderate the relationship (interaction $p = 0.10$).

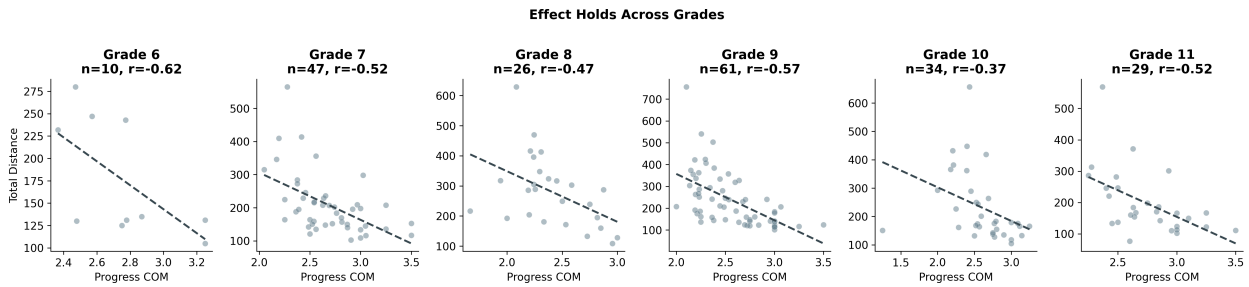

Figure 8: The effect holds across grades 6 to 11. Grades 3–5 and 12 ($n = 2$–4 each) were included in main analyses but omitted from this visualization due to small samples.

4 Discussion

4.1 Why Does Progress Timing Matter?

We interpret progress timing as reflecting different solving approaches. Students whose progress occurs early are completing constraints as they go, committing to solutions quickly. Students whose progress occurs late are exploring first, deferring constraint completion until they have more information.

The greedy approach is risky. When early commitments happen to be correct, the solver finishes quickly. When they are wrong, the solver must backtrack extensively, sometimes traversing the entire grid to undo and redo work. This explains the high variance among greedy solvers: some get lucky, others get stuck.

The distributed approach is safer. By keeping constraint satisfaction low until late in the attempt, the solver retains flexibility. They can adjust their emerging solution as they gather information from other parts of the grid. When they do commit, their placements are more likely to be correct.

4.2 Correction Timing as a Consequence

The negative correlation between Progress COM and Correction COM suggests that these behaviors are linked. Students whose progress occurs late, by virtue of working across the grid, are positioned to notice and correct errors locally. Students whose progress occurs early may not detect conflicts until later, when corrections require traversing back across completed regions.

4.3 Limitations

This study is correlational. We cannot conclude that adopting a distributed approach causes better efficiency. More skilled solvers may naturally gravitate toward distributed strategies, or some third factor may drive both strategy choice and outcomes.

Progress COM measures when progress occurs, not strategy directly. Two students with identical Progress COM could be using different approaches. The variance findings support this interpretation: some greedy solvers do very well, implying that early commitment is not inherently problematic, only when it leads to conflicts.

We tested whether correction concentration moderated the main effect using ordinary least squares (OLS) regression with an interaction term (Progress COM $\times$ Gini coefficient). The interaction was not significant ($\beta = 163$, $p = 0.53$), indicating that the relationship between progress timing and efficiency does not depend on how clustered corrections are.

5 Conclusion

We analyzed 221 puzzle-solving attempts and found that when constraint satisfaction progress occurs strongly predicts efficiency. Students whose progress occurred later traveled substantially less total distance ($r = -0.52$, $d = 1.14$). This advantage was robust across puzzles and grade levels.

Critically, early progress did not uniformly lead to poor outcomes. Instead, greedy solvers showed high variance: some finished efficiently, others struggled badly. Distributed solvers were consistently efficient.

The core insight: staying flexible reduces the need to maneuver.

References

Batenburg, K. J., & Kosters, W. A. (2012). On the difficulty of Nonograms. *ICGA Journal*, 35(4), 195–205.